Introduction to Statistics 02402/02323

Lecture 11: One-way Analysis of Variance, ANOVA

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Overview

1. Summary
2. Introduction
3. Model and Hypotheses
4. Computation: Variance Decomposition and ANOVA Table
   - Estimation of parametre
   - Variability and relation with the t-test for two samples
5. Hypothesis Testing (F-test)
6. Post hoc comparisons
   - Confidence interval for difference between two groups
   - Post hoc t-test for pair-wise difference between groups
7. Model Control
8. A Worked Example – From the Book

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Summary - The general linear model

The last couple of times, we have talked about the *Linear Regression Model*:

$$Y_i = \beta_0 + \beta_1 x_{1,i} + \ldots + \beta_p x_{p,i} + \varepsilon_i$$

- Where $x_{1i}, x_{2i}, \ldots, x_{pi}$ is *quantitative explanatory variables*.
- We assume $\varepsilon_i \sim N(0, \sigma^2)$ (and i.i.d.).

We have also (earlier in the course) talked about a statistical model for 2 groups:

$$Y_i = \mu_1 + \delta x_i + \varepsilon_i$$

- here x_i is *binary (0/1) explanatory variable*.
- Here it is assumed $\varepsilon_i \sim N(0, \sigma^2)$ (and i.i.d.), that is, this is the "pooled" example with equal variance.

Analysis of Variance - ANOVA

"ANalysis Of VAriance" (ANOVA) was introduced by R.A. Fisher about 100 years ago as a systematic way to analyze groups and has since been fundamental to the development of statistics.

- Today: A single classification criterion (one-way ANOVA)
- *In course 02402, also covers: Two classification criteria (two-way ANOVA)*
- Classification criterion = factor
- The first factor is typically called *treatment*, the second factor *block*

Fisher's Terminology: Treatment and Block

Treatment

- A specific condition or intervention applied to experimental units.
- Often corresponds to different levels or types of a factor (e.g. different fertilizers, teaching methods, or dosages).
- **Example:** In an agricultural experiment, applying *Fertilizer A*, *Fertilizer B*, and *No Fertilizer* are three different treatments.

Block

- A group of experimental units that are relatively homogeneous. Blocks are used to control for known sources of variation (e.g. field location, time of measurement, classroom).
- Within each block, treatments are randomly assigned.
- **Example:** A field divided into *North*, *Central*, and *South* zones—each zone is a block because soil quality differs.

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One-Way Analysis of Variance - Example

Data with a qualitative explanatory variable.

Group A	Group B	Group C
2.8	5.5	5.8
3.6	6.3	8.3
3.4	6.1	6.9
2.3	5.7	6.1

Is there a difference (in mean) between groups A, B, and C?

Analysis of variance (ANOVA) can be used for the analysis, provided the observations in each group can be assumed to be normally distributed.

Example in Python

- Go to today's Python notebook in VS Code
  - "Example: Intro to ANOVA"

Vevox Q1,2,3



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One-Way Analysis of Variance - Model

- The model can be written as

$$Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}\,,$$

where it is assumed that ε_{ij} are i.i.d. with

$$\varepsilon_{ij} \sim N(0, \sigma^2)\,.$$

- μ is the overall mean
- α_i indicates the effect of group (treatment) i
- Y_{ij} is measurement j in group i (j ranges from 1 to n_i)

One-Way Analysis of Variance - Hypothesis Test

- We will now compare (more than two) means $(\mu + \alpha_i)$ in the model

$$Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}, \quad \varepsilon_{ij} \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2).$$

- The null hypothesis is given by:

$$H_0: \quad \alpha_i = 0 \quad \text{for all } i.$$

- The alternative hypothesis is given by:

$$H_1: \quad \alpha_i \neq 0 \quad \text{for at least one } i.$$

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One-Way ANOVA - Variance Decomposition

With the model

$$Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}, \quad \varepsilon_{ij} \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2)$$

the total variation in data can be decomposed:

$$SST = SS(Tr) + SSE\,.$$

- SST: "Total variation"
- SS(Tr): Variation *between* the groups
- SSE: Variation *within* the groups
- 'One-way' implies that there is only one factor in the experiment (with k levels).
- analysis of variance means testing is done by comparing variances.

Formulas for Sum of Squares

- Total variation:

$$SST = \sum_{i=1}^{k} \sum_{j=1}^{n_i} (y_{ij} - \bar{y})^2$$

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- Variation within groups (Residual variation left after the model):

$$SSE = \sum_{i=1}^{k} \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_i)^2$$

Formulas for Sum of Squares

- Total variation:

$$SST = \sum_{i=1}^{k} \sum_{j=1}^{n_i} (y_{ij} - \bar{y})^2$$

- Variation within groups (Residual variation left after the model):

$$SSE = \sum_{i=1}^{k} \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_i)^2$$

- Variation between groups (Variation explained by the model):

$$SS(Tr) = \sum_{i=1}^{k} n_i (\bar{y}_i - \bar{y})^2$$

One-Way ANOVA - Parameter Estimates

$$Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}, \quad \varepsilon_{ij} \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2)$$

- $\hat{\mu} = \bar{y}$
- $\hat{\alpha}_i = \bar{y}_i - \bar{y}$
- $\hat{\sigma}^2 = MSE = \frac{SSE}{n-k}$

Vevox Q4



Visual Studio Code

(μ, α_i and σ^2)

Decomposition of variation in data

Theorem 8.2 Variability decomposition

The total sum of squares (SST) can be decomposed into sum of squared errors (SSE) and treatment sum of squares ($SS(Tr)$)

$$\underbrace{\sum_{i=1}^{k}\sum_{j=1}^{n_i}(y_{ij}-\bar{y})^2}_{\text{SST}} = \underbrace{\sum_{i=1}^{k}\sum_{j=1}^{n_i}(y_{ij}-\bar{y}_i)^2}_{\text{SSE}} + \underbrace{\sum_{i=1}^{k} n_i(\bar{y}_i-\bar{y})^2}_{\text{SS(Tr)}}, \tag{8-6}$$

where

$$\bar{y} = \frac{1}{n}\sum_{j=1}^{k}\sum_{j=1}^{n_i} y_{ij}, \quad \bar{y}_i = \frac{1}{n_i}\sum_{j=1}^{n_i} y_{ij}. \tag{8-7}$$

Expressed in short form

$$SST = SS(Tr) + SSE. \tag{8-8}$$

Estimation of parameters

$$Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}, \quad \varepsilon_{ij} \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2)$$

- $\hat{\mu} = \bar{y}$
- $\hat{\alpha}_i = \bar{y}_i - \bar{y}$
- $\hat{\sigma}^2 = \text{"}MSE\text{"} = \frac{SSE}{n-k}$

What is $\bar{y}$:

_____

What is $\bar{y}_i$:

_____

- Go to Python notebook in VS Code
  2) ANOVA: Estimate parameters μ, α_i and σ^2



MSE - pooled variance within the groups

Theorem 8.4 Within group variability

The sum of squared errors SSE divided by $n-k$, also called the residual mean square $MSE = SSE/(n-k)$ is the weighted average of the sample variances from each group

$$MSE = \frac{SSE}{n-k} = \frac{(n_1-1)s_1^2 + \cdots + (n_k-1)s_k^2}{n-k}, \tag{8-14}$$

where s_i^2 is the variance within the ith group

$$s_i^2 = \frac{1}{n_i - 1}\sum_{j=1}^{n_i}(y_{ij} - \bar{y}_i)^2. \tag{8-15}$$

When $k = 2$, that is, we are in the two-sample case presented in Section 3.2, the result here is a copy of the pooled variance expression in Method 3.52

$$\text{For } k = 2: \; MSE = s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n-2}. \tag{8-16}$$

Variance Analysis : ANOVA Table

Source of variation	Deg. of freedom	Sums of squares	Mean sum of squares
Treatment	$k-1$	$SS(Tr)$	$MS(Tr) = \frac{SS(Tr)}{k-1}$
Residual	$n-k$	SSE	$MSE = \frac{SSE}{n-k}$
Total	$n-1$	SST	

n: Number of observations
k: Number of groups

- Go to Python notebook in VS Code

  3) ANOVA: The ANOVA table with Python "ols"



Visual Studio Code

(ANOVA table)

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One-Way ANOVA - F-Test

- We have (Theorem 8.2)

$$SST = SS(Tr) + SSE$$

- From this, the test statistic can be derived:

$$F = \frac{SS(Tr)/(k-1)}{SSE/(n-k)} = \frac{MS(Tr)}{MSE} = \frac{\text{"between group variation"}}{\text{"within group variation"}},$$

where

  - k is the number of groups,
  - n is the number of observations.

What would one conclude when F is very large/small?

F-test

Under the null hypothesis, F follows an F-distribution:

Theorem 8.6

Under the null hypothesis

$$H_0: \quad \alpha_i = 0, \quad i = 1, 2, \ldots, k, \tag{8-18}$$

the test statistic

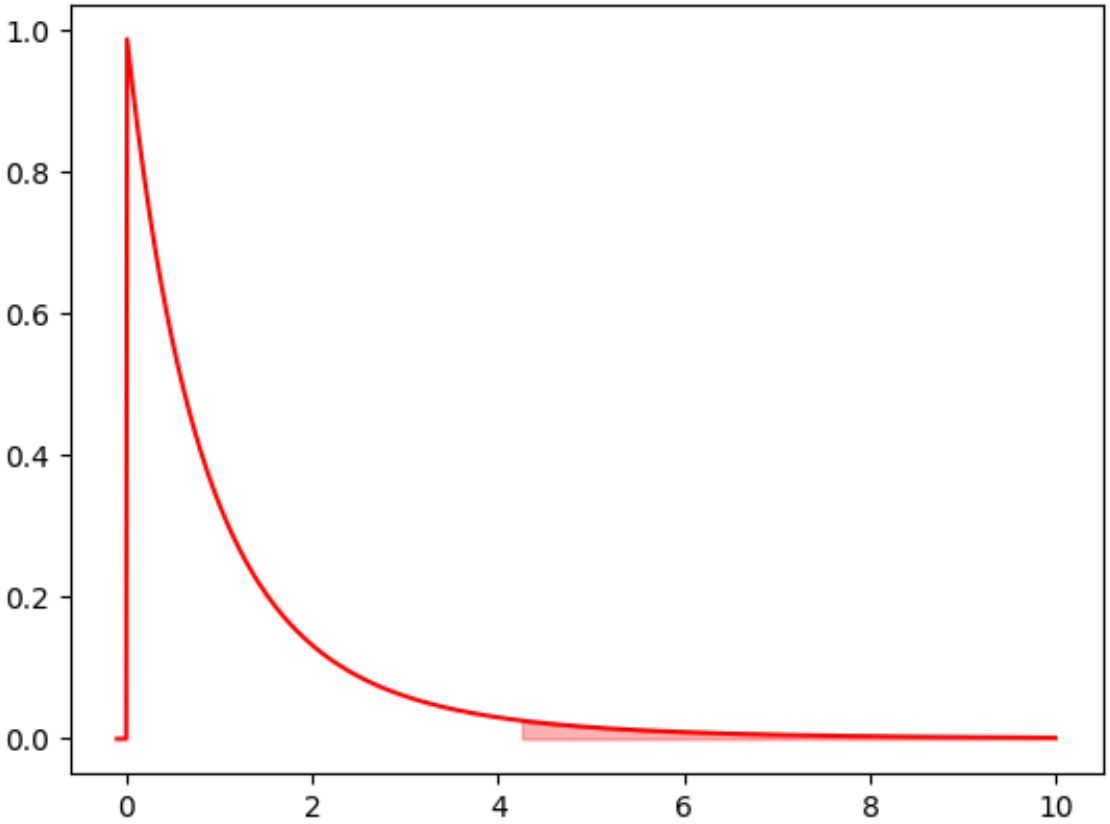
$$F = \frac{SS(Tr)/(k-1)}{SSE/(n-k)}, \tag{8-19}$$

follows an F-distribution with $k-1$ and $n-k$ degrees of freedom.

See Exercise 2.97 + the Appendix in the book about the F-distribution.

F Distribution and F-Test

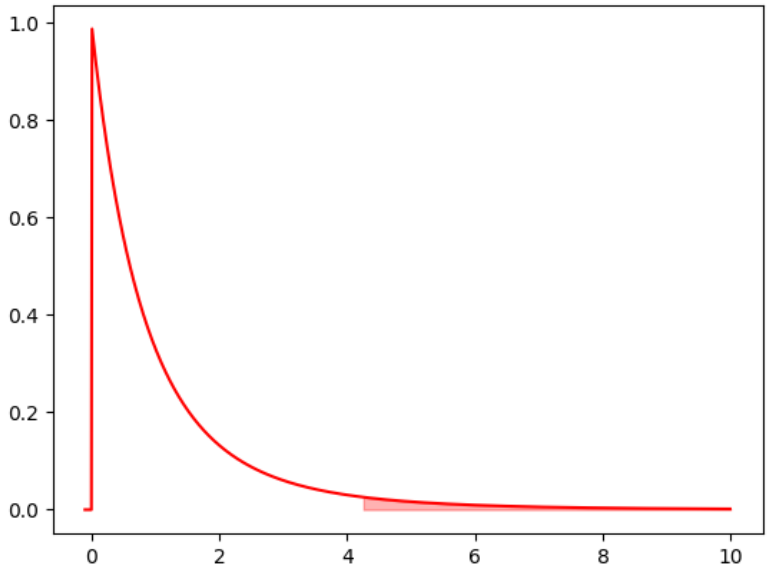
Plot of F-distribution with $n = 12$ and $k = 3$:



The 5% *most extreme* values are above 4.26.

Thinking break (draw and tell)

The F-test is a one-sided test!



$$F = \frac{SS(Tr)/(k-1)}{SSE/(n-k)} = \frac{MS(Tr)}{MSE}$$

What is "F_{obs}" and "F_{crit}"?

How do we find the p-value? (and how do you do it in Python?)

Analysis of variance table

Source of variation	Deg. of freedom	Sums of squares	Mean sum of squares	Test-statistic F	p-value
treatment	$k-1$	$SS(Tr)$	$MS(Tr) = \frac{SS(Tr)}{k-1}$	$F_{\text{obs}} = \frac{MS(Tr)}{MSE}$	$P(F > F_{\text{obs}})$
Residual	$n-k$	SSE	$MSE = \frac{SSE}{n-k}$		
Total	$n-1$	SST			

- Go to Python notebook in VS Code
  - Example: F-test (+VEVOX 5, 6, 7, 8)



(F-test)

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Why is Post-hoc Comparison Necessary?

- ANOVA tests the overall null hypothesis that all group means are equal $H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$.
- If ANOVA shows a significant result (i.e., $p - value < \alpha$), it only indicates that at least one group mean differs, but it doesn't specify which groups are different from each other.
- Post-hoc comparisons are performed to pinpoint which specific group means are different.

Post hoc confidence interval for pairwise difference between groups

Method 8.9 Post hoc pairwise confidence intervals

A single pre-planned $(1-\alpha)\cdot 100\%$ confidence interval for the difference between treatment i and j is found as

$$\bar{y}_i - \bar{y}_j \pm t_{1-\alpha/2}\sqrt{MSE\left(\frac{1}{n_i}+\frac{1}{n_j}\right)}, \tag{8-22}$$

where $t_{1-\alpha/2}$ is based on the t-distribution with $n-k$ degrees of freedom.

If all $M = k(k-1)/2$ combinations of pairwise confidence intervals are calculated using the formula M times, but each time with $\alpha_{\text{Bonferroni}} = \alpha/M$ (see Remark 8.14 below).

Revisit also Theorem 3.54

Note the fewer degrees of freedom, since more parameters are estimated in the calculation of $MSE = SSE/(n-k) = s_p^2$ (the pooled variance estimate).

Post hoc calculation of "Least Significant Difference"

Remark 8.13 Least Significant Difference (LSD) values

If there is the same number of observations in each treatment group $m = n_1 = \ldots = n_k$ the LSD value for a particular significance level

$$LSD_\alpha = t_{1-\alpha/2}\sqrt{2 \cdot MSE/m} \tag{8-28}$$

will have the same value for all the possible comparisons made.
The LSD value is particularly useful as a "measuring stick" with which we can go and compare all the observed means directly: the observed means with difference higher than the LSD are significantly different on the α-level. When used for all of the comparisons, as suggested, one should as level use the Bonferroni corrected version $LSD_{\alpha_{\text{Bonferroni}}}$ (see Remark 8.14 below for an elaborated explanation).

Post hoc pairwise hypothesis tests

Method 8.10 Post hoc pairwise hypothesis tests

A single pre-planned level α hypothesis tests

$$H_0: \ \mu_i = \mu_j, \quad H_1: \ \mu_i \neq \mu_j, \tag{8-23}$$

is carried out by

$$t_{\text{obs}} = \frac{\bar{y}_i - \bar{y}_j}{\sqrt{MSE\left(\frac{1}{n_i} + \frac{1}{n_j}\right)}}, \tag{8-24}$$

and

$$p\text{-value} = 2 \cdot P(T > |t_{\text{obs}}|), \tag{8-25}$$

where the t-distribution with $n - k$ degrees of freedom is used.

If all $M = k(k-1)/2$ combinations of pairwise hypothesis tests are carried out use the approach M times but each time with test level $\alpha_{\text{Bonferroni}} = \alpha / M$ (see Remark 8.14 below).

Bonferrerroni correction of α

Did you grasp the idea of the "Bonferroni correction" of α?

What do you do, and why do you do it?

(Read section 8.14 in the book yourself)

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Model control (analysis of residuals)

Method 8.15 Normality control in one-way ANOVA

To control for the normality assumptions in one-way ANOVA we perform a q-q plot on the pooled set of n estimated residuals

$$e_{ij} = y_{ij} - \bar{y}_i,\ j = 1, \ldots, n_i,\ i = 1 \ldots, k. \tag{8-32}$$

Model Control

Our model:

$$Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}, \quad \varepsilon_{ij} \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2).$$

What is our assumption(s)?
Can we do anything to check if the assumptions are valid?

- Use a q-q plot on residuals to check for the normality assumption
- Check variance homogeneity by categorized box plots

VEVOX Q9



Visual Studio Code
(Model control)

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8. A Worked Example – From the Book

A Worked Example – From the Book

8.2.5 A complete worked through example: plastic types for lamps

Example 8.17 Plastic types for lamps

On a lamp two plastic screens are to be mounted. It is essential that these plastic screens have a good impact strength. Therefore an experiment is carried out for 5 different types of plastic. 6 samples in each plastic type are tested. The strengths of these items are determined. The following measurement data was found (strength in kJ/m^2):

Type of plastic				
I	II	III	IV	V
44.6	52.8	53.1	51.5	48.2
50.5	58.3	50.0	53.7	40.8
46.3	55.4	54.4	50.5	44.5
48.5	57.4	55.3	54.4	43.9
45.2	58.1	50.6	47.5	45.9
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7. Model Control
8. A Worked Example – From the Book